

		Force by 2kg = 3170
6	D	For elastic collision, momentum and total KE must be conserved. Only option D fulfilled both the conditions. For Option D: Initial total momentum = $(0.50)(4.0) + (0.50)(1.0) = 2.5 \text{ N s}^{-1}$ Final total momentum = $(0.50)(1.0) + (0.50)(4.0) = 2.5 \text{ N s}^{-1}$ Initial total KE = $\frac{1}{2}(0.50)(4.0)^2 + \frac{1}{2}(0.50)(1.0)^2 = 4.25 \text{ J}$ Final total KE = $\frac{1}{2}(0.50)(1.0)^2 + \frac{1}{2}(0.50)(4.0)^2 = 4.25 \text{ J}$ Extra knowledge: since the object has the same mass, and it is head on elastic collision, the object will swap their velocities before and after collision
7	C	Weight = (Tension) $\cos(20^\circ)$